Little Sub and Triples

Tags:

Time Limit: 1 s Memory Limit: 256 MB
Submission: 1485 AC: 258 Score: 88.36

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Description

Little Sub has learned a new word 'Triple', which usually means a group with three elements in it. Now he comes up with an interesting problem for you.

Define F(a,b,c) = e^a□e^b / e^c

Given an positive integer sequence A and some queries. In each query, you have to tell if there exists a triple T = (x, y, z) such that:

1.x, y, z are unique integers.

2.l ≤ x, y, z ≤ r.

3.Ax ≤ Ay ≤ Az.

4.F(Ax,Ay,Az) > 1.

Input

The first line contains two positive integers n, q(1 ≤ n, q ≤ 200000).
The second line contains n positive integers, indicating the integer sequence. The next q lines describe each query by giving two integer l, r(1 ≤ l ≤ r ≤ n). All given integers will not exceed 2^31 − 1.

Due to some reasons, when l>r, please consider it as an empty interval. Sorry for it.

Output

For each query, print YES if it exists any legal triple or NO otherwise.

Samples

input Copy

```
4 2
1 2 3 4
1 3
2 4
```

output Copy

```
NO
YES
```

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